

Lecture 6

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Convention. The ground set is $E = \{e_1, \dots, e_n\}$. For $e \notin S$, write

$$f(e \mid S) = f(S + e) - f(S),$$

and let $[a]_+ = \max\{a, 0\}$. Unless stated otherwise, $f : 2^E \rightarrow \mathbb{R}_{\geq 0}$ is submodular but need not be monotone or normalized. For vectors, $x \vee y$ denotes the coordinatewise maximum.

1 Double Greedy

We consider unconstrained nonmonotone submodular maximization

$$\max\{f(S) : S \subseteq E\}. \quad (1)$$

Following the complementary-set notation in the handwritten notes, define

$$\bar{f}(S) = f(E \setminus S).$$

The algorithm maintains two disjoint sets: A_i contains the elements already accepted, while $\bar{B}_i$ contains the elements already rejected. Write $B_i = E \setminus \bar{B}_i$. At the start of iteration i , the element e_i belongs to neither A_{i-1} nor $\bar{B}_{i-1}$, and $A_{i-1} \subseteq B_{i-1} - e_i$.

Define the raw gains

$$a_i = f(e_i \mid A_{i-1}), \quad b_i = \bar{f}(e_i \mid \bar{B}_{i-1}) = f(B_{i-1} - e_i) - f(B_{i-1}), \quad (2)$$

By diminishing returns,

$$a_i + b_i = f(e_i \mid A_{i-1}) - f(e_i \mid B_{i-1} - e_i) \geq 0. \quad (3)$$

1.1 Deterministic double greedy

Algorithm 1 Deterministic double greedy

Input: A value oracle for f and an ordering $E = (e_1, \dots, e_n)$.

Initialize: $A_0 \leftarrow \emptyset$ and $\bar{B}_0 \leftarrow \emptyset$.

for $i = 1, \dots, n$ **do**

$a_i \leftarrow f(e_i \mid A_{i-1})$ and $b_i \leftarrow \bar{f}(e_i \mid \bar{B}_{i-1})$.

if $a_i \geq b_i$ **then**

$A_i \leftarrow A_{i-1} + e_i$ and $\bar{B}_i \leftarrow \bar{B}_{i-1}$.

else

$A_i \leftarrow A_{i-1}$ and $\bar{B}_i \leftarrow \bar{B}_{i-1} + e_i$.

end if

end for

return A_n ; at termination, $A_n = E \setminus \bar{B}_n$.

Theorem 1.1 (Deterministic double greedy). *Deterministic double greedy is a $1/3$ -approximation [3].*

1.2 Randomized double greedy

For the randomized version, truncate the two gains at zero:

$$\alpha_i = [a_i]_+, \quad \beta_i = [b_i]_+.$$

Algorithm 2 Randomized double greedy

Input: A value oracle for f and an ordering $E = (e_1, \dots, e_n)$.

Initialize: $A_0 \leftarrow \emptyset$ and $\bar{B}_0 \leftarrow \emptyset$.

for $i = 1, \dots, n$ **do**

$a_i \leftarrow f(e_i \mid A_{i-1})$ and $b_i \leftarrow \bar{f}(e_i \mid \bar{B}_{i-1})$.

$\alpha_i \leftarrow [a_i]_+$ and $\beta_i \leftarrow [b_i]_+$.

if $\alpha_i + \beta_i = 0$ **then**

$p_i \leftarrow 1$ and $q_i \leftarrow 0$.

else

$p_i \leftarrow \alpha_i / (\alpha_i + \beta_i)$ and $q_i \leftarrow \beta_i / (\alpha_i + \beta_i)$.

end if

with probability p_i : $A_i \leftarrow A_{i-1} + e_i$ and $\bar{B}_i \leftarrow \bar{B}_{i-1}$.

with probability q_i : $A_i \leftarrow A_{i-1}$ and $\bar{B}_i \leftarrow \bar{B}_{i-1} + e_i$.

end for

return A_n ; at termination, $A_n = E \setminus \bar{B}_n$.

For the analysis, define the hybrid set

$$O_i = (\text{OPT} \cup A_i) \setminus \bar{B}_i. \quad (4)$$

Thus $O_0 = \text{OPT}$, $O_n = A_n$, and $A_i \subseteq O_i \subseteq B_i$. Let $\mathbb{E}_i$ denote expectation over the choice at iteration i , conditioned on all earlier choices.

Lemma 1.2 (One-step comparison). *At every iteration,*

$$\begin{aligned} \mathbb{E}_i[f(O_{i-1}) - f(O_i)] &\leq \frac{1}{2} \mathbb{E}_i[f(A_i) - f(A_{i-1}) \\ &\quad + \bar{f}(\bar{B}_i) - \bar{f}(\bar{B}_{i-1})]. \end{aligned}$$

Proof. First suppose $\alpha_i + \beta_i > 0$. Conditioned on A_{i-1} and $\bar{B}_{i-1}$, the expected increase in the two objectives is

$$p_i a_i + q_i b_i = \frac{\alpha_i^2 + \beta_i^2}{\alpha_i + \beta_i}. \quad (5)$$

Indeed, a branch whose raw gain is negative has probability zero.

If $e_i \notin \text{OPT}$, then O_i changes only when e_i is added to A_i . Since $O_{i-1} \subseteq B_{i-1} - e_i$, the loss in that branch satisfies

$$f(O_{i-1}) - f(O_{i-1} + e_i) = -f(e_i \mid O_{i-1}) \leq -f(e_i \mid B_{i-1} - e_i) = b_i \leq \beta_i.$$

This branch has probability p_i , so the expected hybrid loss is at most $p_i \beta_i$. If $e_i \in \text{OPT}$, then O_i changes only when e_i is added to $\bar{B}_i$. Since $A_{i-1} \subseteq O_{i-1} - e_i$, the loss in that branch satisfies

$$f(O_{i-1}) - f(O_{i-1} - e_i) = f(e_i \mid O_{i-1} - e_i) \leq f(e_i \mid A_{i-1}) = a_i \leq \alpha_i.$$

This branch has probability q_i , so its expected loss is at most $q_i\alpha_i$. In either case,

$$\mathbb{E}_i[f(O_{i-1}) - f(O_i)] \leq \frac{\alpha_i\beta_i}{\alpha_i + \beta_i} \leq \frac{1}{2} \frac{\alpha_i^2 + \beta_i^2}{\alpha_i + \beta_i},$$

where the last inequality is $2\alpha_i\beta_i \leq \alpha_i^2 + \beta_i^2$.

If $\alpha_i + \beta_i = 0$, then $a_i, b_i \leq 0$. Together with (3), this implies $a_i = b_i = 0$. The algorithm adds e_i to A_i . The two-objective gain is zero; if $e_i \in \text{OPT}$, then the hybrid set does not change, while if $e_i \notin \text{OPT}$, its loss is at most $b_i = 0$. Hence the lemma also holds in this case. $\square$

Theorem 1.3 (Randomized double greedy). *Randomized double greedy is a $1/2$ -approximation:*

$$\mathbb{E}[f(A_n)] \geq \frac{1}{2}f(\text{OPT}).$$

Proof. Take expectations in Lemma 1.2 and sum over all iterations. Since $O_0 = \text{OPT}$, $O_n = A_n$, $\bar{f}(\bar{B}_n) = f(A_n)$, and $\bar{f}(\emptyset) = f(E)$, we obtain

$$\begin{aligned} f(\text{OPT}) - \mathbb{E}[f(A_n)] &\leq \frac{1}{2}\mathbb{E}[f(A_n) - f(\emptyset) + \bar{f}(\bar{B}_n) - \bar{f}(\emptyset)] \\ &= \mathbb{E}[f(A_n)] - \frac{1}{2}(f(\emptyset) + f(E)) \\ &\leq \mathbb{E}[f(A_n)]. \end{aligned}$$

The last inequality uses $f \geq 0$, and rearranging proves the result [3]. $\square$

Randomized double greedy makes only a linear number of value-oracle queries. The $1/2$ factor cannot be improved by a polynomial-query value-oracle algorithm in general [2].

2 Continuous Greedy

We now return to the multilinear extension F defined earlier. If f is monotone, the first-derivative identity implies $\nabla F(x) \geq 0$, so F is coordinatewise monotone. Concavity along nonnegative directions also gives the tangent inequality

$$F(y) - F(x) \leq \langle \nabla F(x), y - x \rangle \quad \text{whenever } 0 \leq x \leq y \leq \mathbf{1}. \quad (6)$$

Let $P \subseteq [0, 1]^E$ be a nonempty, convex, down-closed polytope over which linear functions can be optimized. We compare against an integral optimal solution:

$$\mathbf{1}_{\text{OPT}} \in P, \quad f(\text{OPT}) = \max\{f(S) : \mathbf{1}_S \in P\}.$$

Algorithm 3 Ideal continuous greedy

Input: The multilinear extension F with exact gradient access;

a down-closed polytope P with exact linear optimization; and $T \in [0, 1]$.

Initialize: $x(0) \leftarrow 0$.

for continuous time $t \in [0, T]$ **do**

Choose, with a fixed measurable tie-breaking rule, $v(t) \in \arg \max_{v \in P} \langle v, \nabla F(x(t)) \rangle$.

Evolve x according to $\dot{x}(t) = v(t)$ almost everywhere.

end for

return $x(T)$.

Algorithm 3 is an ideal continuous-time process, rather than a finite oracle algorithm; discretization is discussed in Section 6.

The trajectory is feasible. Indeed, for $t > 0$,

$$\frac{x(t)}{t} = \frac{1}{t} \int_0^t v(s) ds \in P$$

by convexity. Since $t \leq 1$ and P is down-closed, $x(t) \in P$. More explicitly, $x(t) = t(x(t)/t) \leq x(t)/t$ coordinatewise.

Theorem 2.1 (Continuous greedy). *If f is monotone, the ideal process with $T = 1$ satisfies*

$$F(x(1)) \geq \left(1 - \frac{1}{e}\right) f(\text{OPT}).$$

Proof. Let $x^* = \mathbf{1}_{\text{OPT}}$ and define

$$d(t) = (x^* - x(t))_+ = x(t) \vee x^* - x(t).$$

Since $0 \leq d(t) \leq x^*$ and P is down-closed, $d(t) \in P$. Optimality of $v(t)$ and (6) give

$$\begin{aligned} \frac{d}{dt} F(x(t)) &= \langle \nabla F(x(t)), v(t) \rangle \\ &\geq \langle \nabla F(x(t)), d(t) \rangle \\ &\geq F(x(t) \vee x^*) - F(x(t)) \\ &\geq f(\text{OPT}) - F(x(t)), \end{aligned}$$

where the last step uses monotonicity. Multiplying by e^t and integrating from 0 to t yields

$$F(x(t)) \geq (1 - e^{-t})f(\text{OPT}) + e^{-t}F(0) \geq (1 - e^{-t})f(\text{OPT}).$$

The result follows by setting $t = 1$ [4]. □

2.1 A nonmonotone stopping-time bound

For a nonmonotone objective, the last inequality in the preceding proof is unavailable. We replace it by the following estimate, used in analyses of measured continuous greedy [5].

Lemma 2.2 (Union lower bound). *Suppose f is nonnegative and submodular. If $x \in [0, 1]^E$ and $\|x\|_\infty \leq p \leq 1$, then every $S \subseteq E$ satisfies*

$$F(x \vee \mathbf{1}_S) \geq (1 - p)f(S). \tag{7}$$

Proof. The case $p = 0$ is immediate. Assume $p > 0$, set $u = (x \vee \mathbf{1}_S) - \mathbf{1}_S$, and define

$$g(\lambda) = F(\mathbf{1}_S + \lambda u), \quad 0 \leq \lambda \leq 1/p.$$

We have $0 \leq u \leq p(\mathbf{1} - \mathbf{1}_S)$, so the entire displayed segment stays in $[0, 1]^E$. The vector u is nonnegative, so g is concave. Since $1 = (1 - p) \cdot 0 + p \cdot (1/p)$,

$$F(x \vee \mathbf{1}_S) = g(1) \geq (1 - p)g(0) + pg(1/p) \geq (1 - p)f(S).$$

The final inequality uses $F \geq 0$. □

Theorem 2.3 (Early-stopped continuous greedy). *For a nonnegative, possibly nonmonotone function f , the ideal continuous-greedy process stopped at $T = \ln 2$ satisfies*

$$F(x(T)) \geq (1 - \ln 2)f(\text{OPT}).$$

Proof. Since $v(t) \in [0, 1]^E$, each coordinate satisfies $x_e(t) \leq t$. The first three lines in the proof of the monotone guarantee remain valid. Lemma 2.2, applied with $p = t$, therefore gives

$$\frac{d}{dt}F(x(t)) \geq (1 - t)f(\text{OPT}) - F(x(t)).$$

Multiplying by e^t and integrating gives

$$F(x(T)) \geq (2 - T - 2e^{-T})f(\text{OPT}) + e^{-T}F(0).$$

The coefficient $2 - T - 2e^{-T}$ is maximized on $[0, 1]$ at $T = \ln 2$, where it equals $1 - \ln 2$. $\square$

3 Measured Continuous Greedy

We use the capped-direction formulation appearing in the handwritten notes. At a point x , the direction must belong to P and must fit inside the remaining coordinatewise capacity $\mathbf{1} - x$.

Algorithm 4 Ideal measured continuous greedy

Input: The multilinear extension F with exact gradient access;

a down-closed polytope P with exact optimization over $\{v \in P : v \leq \mathbf{1} - x\}$; and $T \in [0, 1]$.

Initialize: $x(0) \leftarrow 0$.

for continuous time $t \in [0, T]$ **do**

Choose, with a fixed measurable tie-breaking rule,

$v(t) \in \arg \max \{ \langle v, \nabla F(x(t)) \rangle : v \in P, v \leq \mathbf{1} - x(t) \}.$

Evolve x according to $\dot{x}(t) = v(t)$ almost everywhere.

end for

return $x(T)$.

The feasible region is nonempty at every time because it contains 0.

As before, $x(t)/t$ is an average of points in P , so $x(t) \in P$ for $t \leq 1$. The cap in Algorithm 4 also gives

$$\dot{x}_e(t) \leq 1 - x_e(t) \quad \implies \quad x_e(t) \leq 1 - e^{-t}. \quad (8)$$

Theorem 3.1 (Measured continuous greedy). *At time $T \leq 1$, the ideal capped-direction process satisfies*

$$\begin{aligned} F(x(T)) &\geq (1 - e^{-T})f(\text{OPT}) && \text{if } f \text{ is monotone,} \\ F(x(T)) &\geq Te^{-T}f(\text{OPT}) && \text{if } f \text{ is nonnegative and possibly nonmonotone.} \end{aligned}$$

Consequently, at $T = 1$ the guarantees are $1 - 1/e$ and $1/e$, respectively.

Proof. Let $x^* = \mathbf{1}_{\text{OPT}}$ and set

$$d(t) = (x^* - x(t))_+ = x(t) \vee x^* - x(t).$$

We have $d(t) \leq x^* \in P$, so $d(t) \in P$ by down-closedness. Moreover, $d(t) \leq \mathbf{1} - x(t)$. Thus $d(t)$ is a feasible comparator for the direction choice in Algorithm 4, and

$$\begin{aligned} \frac{d}{dt}F(x(t)) &= \langle \nabla F(x(t)), v(t) \rangle \\ &\geq \langle \nabla F(x(t)), d(t) \rangle \\ &\geq F(x(t) \vee x^*) - F(x(t)). \end{aligned} \tag{9}$$

If f is monotone, the right-hand side is at least $f(\text{OPT}) - F(x(t))$, and the continuous-greedy analysis applies. If f is nonmonotone, (8) and Lemma 2.2 imply

$$F(x(t) \vee x^*) \geq e^{-t}f(\text{OPT}).$$

It follows from (9) that

$$\frac{d}{dt}(e^t F(x(t))) \geq f(\text{OPT}).$$

After integration,

$$F(x(T)) \geq e^{-T}F(0) + Te^{-T}f(\text{OPT}) \geq Te^{-T}f(\text{OPT}).$$

This proves the claims. In particular, $1/e \approx 0.3679$ [5]. $\square$

Remark (Oracle requirement). The capped-direction presentation above requires linear optimization over the time-dependent polytope

$$\{v \in P : v \leq \mathbf{1} - x(t)\}.$$

The original measured continuous-greedy algorithm of Feldman, Naor, and Schwartz scales each coordinate of a direction from P by its remaining capacity; that formulation needs only a linear-optimization oracle for P and yields the same ideal differential bounds [5].

4 Discretization and Rounding

The exact constants above describe ideal continuous-time trajectories. A finite implementation uses the relevant linear-optimization oracle, replaces the differential equation by sufficiently small Euler steps, and estimates each coordinate of $\nabla F(x)$ by random sampling. For the capped-direction formulation, the relevant oracle is the stronger one identified in the preceding remark. For every fixed $\varepsilon > 0$, the standard discretized algorithms run in time polynomial in n and $1/\varepsilon$ under these oracles and give, in expectation, $1 - 1/e - \varepsilon$ in the monotone case and $1/e - \varepsilon$ for nonmonotone measured continuous greedy [4, 5].

The output x is fractional. Under a matroid constraint, pipage rounding converts x to an independent set without decreasing the multilinear value in expectation [6, 4]. The relevant geometric fact follows from the mixed partials computed earlier: for distinct i, j , the function

$$t \longmapsto F(x + t(\mathbf{1}_i - \mathbf{1}_j))$$

is convex on every interval for which its argument stays in $[0, 1]^E$. For an arbitrary down-closed polytope, the required rounding procedure is a separate part of the algorithmic problem.

Summary. For unconstrained nonnegative submodular maximization, deterministic double greedy achieves $1/3$ and randomized double greedy achieves the optimal value-oracle factor $1/2$. For a monotone objective over a down-closed polytope, continuous greedy obtains the ideal fractional factor $1 - 1/e$. Ordinary continuous greedy gives $1 - \ln 2$ for a nonmonotone objective when stopped at time $\ln 2$, while measured continuous greedy improves the ideal factor to $1/e$. Discretization, sampling, and rounding turn the continuous trajectories into algorithms for discrete feasible sets.

References

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